MILITARY STANDARD
EXPLOSIVE: SAMPLING, INSPECTION AND TESTING

TO ALL HOLDERS OF MIL-STD-650

1. The following page of MIL-STD-650 have been revised and supersede the pages listed:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
iii	4 September 1987	iii	31 May 1973 (NOTICE 3)
iv	4 September 1987	iv	31 May 1973 (NOTICE 3)
vii	31 May 1973	vii	Reprinted without change
viii	4 September 1987	viii	31 May 1973 (NOTICE 3)

2. New Methods

METHOD NO.	TITLE	DATE
T510.1	Friction Sensitiveness Test	4 September 1987
T511.1	Impact sensitivity Test (ERL, TYPE 12 Apparatus)	4 September 1987
T512.1	Electrostatic Sensitivity Test (ARDEC Laboratory Method)	4 September 1987
T513.1	Large Scale Gap Test	4 September 1987
T514.1	Autoignition Temperature Test	4 September 1987
T515.1	Explosion Temperature Test (ARDEC Laboratory Method)	4 September 1987

3. Retain this notice and insert before the table of contents.

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METHOD T510.1

FRICTION SENSITIVENESS TEST

PICATINNY ARSENAL, NJ

1. SCOPE.

The friction pendulum test is used to determine the sensitiveness of explosives when subjected to a glancing blow with its attendant friction.

2. APPARATUS.

a. Pendulum Friction Device consists of:

(1) Steel swinging shoe (bob of the pendulum)

(2) Frame support of the pendulum (the “A-Frame”)

(3) Two free bases of the "A-Frame" which can be raised or lowered simultaneously, controlling the pendulum swings. Test apparatus is shown diagramatically in Figure 1.

b. The anvil consists of mild steel, Rockwell hardness B71 with a smooth, polished upper face. Three parallel grooves are cut across the central portion to prevent explosive from being bruahes off.

c. Shoes, manufactured from mild steel, Rockwell hardness B71. Hard fiber for the fiber-faced shoe is procured in accordance with ASTM specifics’ ion D-710-43T, commercial grade, density 1.34. It is supplied in three colors: black, red, and gray. It has been indicated that there may be slight differences in material for different colors. To make the test as uniform as possible, only the red is used The fiber is not very resistant, and it is therefore difficult 10 keep this shoe in first-class operating condition under the hard usage to which it is subjected. The condition of the shoe has to be inspected very often.

d. Automatic tripper, used to drop shoe from any desired vertical height ranging from 1/2 to 2 meters. Height drop normally used at Picatinny Arsenal is 1 meter.

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3. PROCEDURE.

3.1 Preparation of Apparatus.

3.1.1 Before trials are begun, test the shoe over the empty anvil to make sure that it is adjusted squarely with the anvil and that when loaded with weights it is adjusted in relation to the anvil so that it will swing across the face of the anvil 18 ± 1 times before coming to rest. Make the adjustment by operating the turnbuckle that raises or lowers the A-Frame.

3.1.2 The temperature of the anvil and shoe shall be controlled to be about 70 ± 5°F.

3.2 Test Procedure

3.2.1 With the device properly adjusted, 7 ± 0.1 grams of the explosive to be tested shall be spread in an even layer in and about the three grooves in the anvil and the shoe dropped upon it. After the pendulum has ceased swinging any remaining explosive shall be brushed off, and the faces of the shoe and anvil shall be cleaned thoroughly with a suitable solvent.

3.2.2 In order to protect himself during cleaning, the operator shall wear a face mask and asbestos gloves. The shoe and anvil shall be rubbed with Carborundum cloth to remove any roughness caused by the impact of the shoe on the soft, steel anvil, also any adherent gritty matter. When the face of the shoe and anvil are clean and dry, another charge of 7 ± 0.1 grams of the explosive shall be placed on the anvil as before.

3.2.3 The observations which have been made and recorded are "explosion" (whether complete or partial), “burning”, (whether complete or partial), "local crackling", (whether distinctly audible or almost inaudible), and "no local crackling”.

3.2.4 The number of swings on which these occurrences take place and the total number of swings for each trial shall be recorded. The number of swings on an empty anvil shall be checked and adjusted after every trial.

3.3 Evaluation and Results.

3.3.1 A test shall consist of ten (10) trials with the steel shoe, except when complete explosion or burning occurs in any trial. If explostion or burning occurs, the trials with the steel shoe shall be discontinued.

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3.3.2 Ten (10) trials shall be made with the fiber faced shoe only when complete explosion or burning occurs with steel shoe, or as prescribed in test directive. If the explosive passes the test with the use of the steel shoe, no further trials shall be conducted. From a viewpoint of economy and efficiency, therefore, all samples shall be tested first with the steel shoe.

3.3.3 An explosive shall also be regarded as passing the friction pendulum test (which is a measure of its sensitiveness to a glancing blow) if in ten (10) trials with the fiber-faced shoe, there is no more than an almost inaudible local crackling, regardless of its behavior when subjected to the action of the steel shoe.

3.3.4 Typical Test Results

Explosive	With Fiber Shoe	With Steel Shoe
PETN	Passed	Failed
RDX	Passed	Failed
TETRYL	Passed	Failed
Composition B	Passed	Passed
TNT	Passed	Passed
EXPLOSIVE D	Passed	Passed

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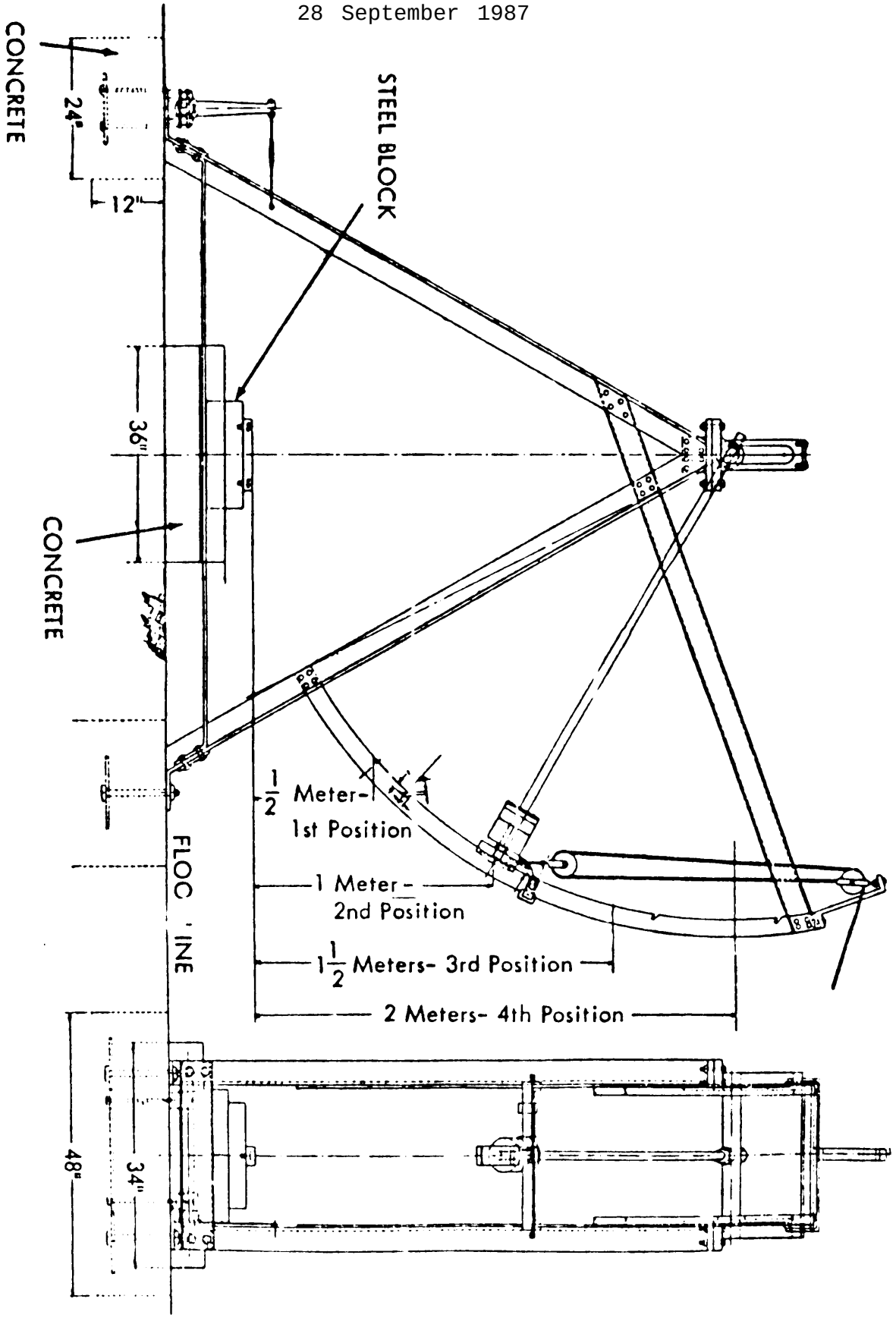


FIG.1 PENDULUM FRICTION DEVICE ASSEMBLY

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METHOD T511.1

IMPACT SENSITIVITY TEST
(ERL, TYPE 12 Apparatus)

1. SCOPE. This test is used to determine the sensitivity of an explosive to impact.

2. APPARATUS.

a. ERL machine (Bruceton Machine), Type 12 tooling

(1) Striker pin, 1.250 in. diameter by 3.5000 in, long, made of tool steel hardened to 60-63 Rockwell "C" scale. Flat surface next to explosive is ground to a finish of 16 u in.

(2) Fine Flint Sandpaper, Norton Co.

(3) Anvil, 1.250 in. diameter by 1.250 in. long, made of tool steel hardened to 60 Rockwell "C", all surfaces ground and polished, mounted in a tool holder assembly which is bolted to the machine base.

b. Weight, 2.5kg made of hardened steel.

3. PROCEDURE.

3.1 Preparation of apparatus.

3.1.1 Figure 1 shows the assembly of the test elements to be used, which consists of a free falling weight, tooling to hold the explosive sample, and a supporting frame.

3.2 Separation of sample.

3.2.1 The test explosives are solid, granular materials which are either pure compounds or mixtures. For materials which are normally cast-loaded into a weapon, prepare by casting as a thin sheet (weight from 3 to 10 g depending on material availability and number 01 determinations to be made).

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3.2.2 Ground the cast sheet gently by hand in a wooden mortar and screen the material through a set of No. 20 and 80 U.S. Standard Sieves. The material that passes through the No. 20 and is retained on the No. 80 screen is used for the test.

3.2.3 Test other solid, granular materials “as recieved” without further pretest processing.

3.3 Test procedure.

3.3.1 Place each test sample of 35 ± 2 mg of explosive in a loose pile in the center of the sandpaper.

3.3.2 Place the sandpaper containing the test sample in the center of the anvil and lower the striker gently so that it rests on the top of the explosive pile.

3.3.3 Elevate the drop weight to a height of 15 cm.

3.3.4 Drop the weight and observe the result.

3.3.5 After each drop, discard the test sample and use a fresh sample for the next drop. Clean the striker and anvil faces which are in contact with the test sample with solvent (such as acetone) after each test.

3.3.6 Replace striker and anvil when working surfaces become roughened as determined by making carbon paper impressions of the surfaces. Old tools are refinished and reused. Replace the striker when its height has diminshed by 0.250 in.

3.4 Evaluation.

3.4.1 A test shall consist of twenty (20) trials.

3.4.2 The candidate material passes the test if in twenty (20) consecutive trials no burning or detonation is observed during impact or in the post test examination of the sample.

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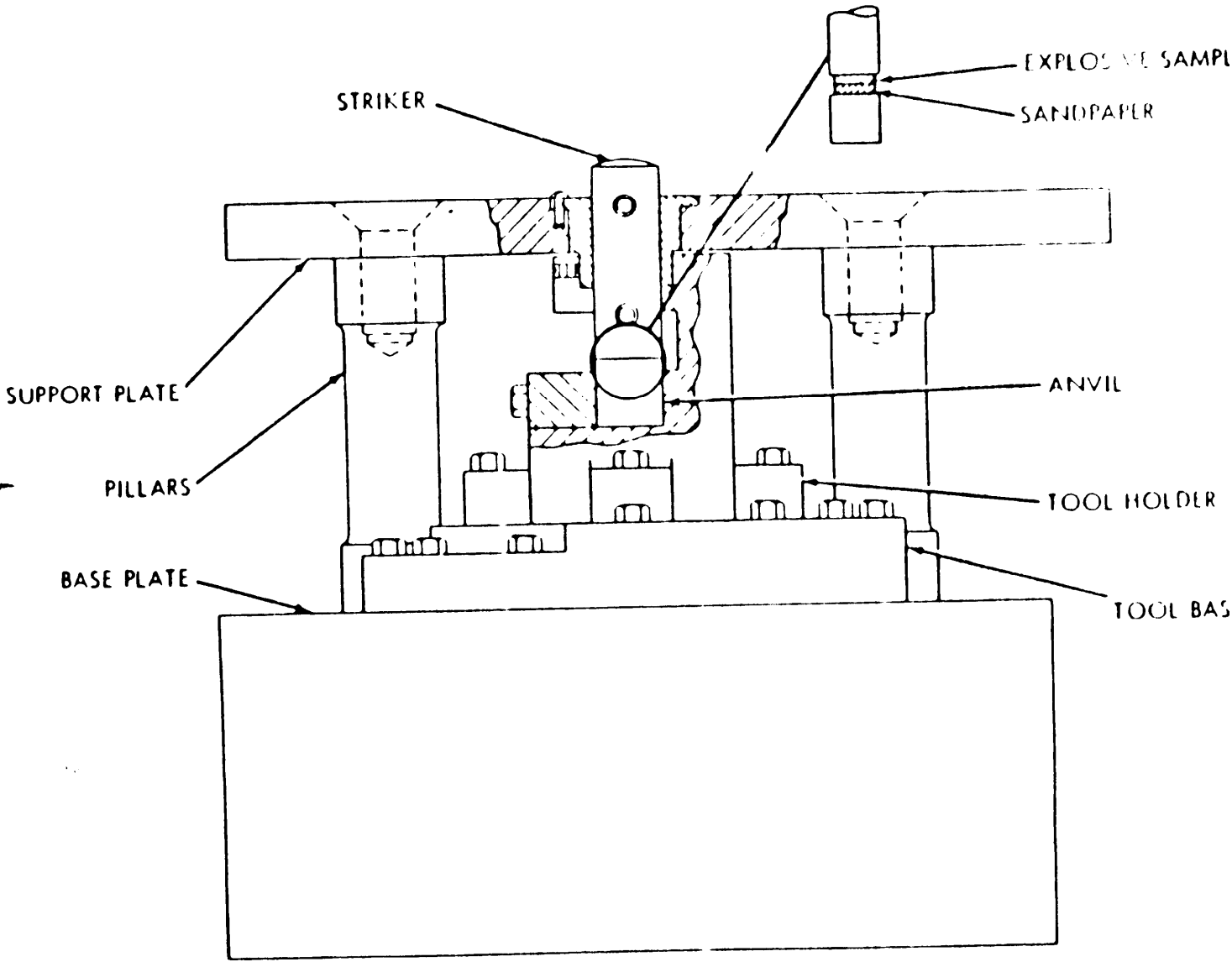


FIG. 1. TOOL HOLDER ASSEMBLY TYPE 12 TOOLS

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METHOD T 512.1

Approaching-Electrode Electrostatic Sensitivity Test (U.S. Army, ARDEC Laboratory Method)

1. Scope. This test is used to determine the electrostatic sensitivity of an energetic material.

2. Apparatus.

a. The approaching-electrode apparatus consists of a 5000 volt variable high voltage power supply, a capacitor-charging, circuit, an electrostatic voltmeter, and approaching-electode assembly.

b. The capacitor-charging circuit consists of low inductance, high voltage capacitor designed so that the appropriate capacitance, from 0.0002 to 0.02 microfarad, could be connected in the circuit, as either a single capacitor or a group, of capacitors in parallel (ref: M.S. Kirshenbaum, “Response of Primary Explosives to Gaseous Discharge in an Improved Approaching-Electrode Electrostatic Sensitivity Apparatus,” PATR 4955, Picatinny Arsenal, N.J. October 1976).

c. The sample holder consists of a 0.9 -1.6 mm thick nylon washer (4.8 mm i.d.) or equivalent, fastened (double adhesive tape may be used) to the top of a 19 mm (3/4 inch) diameter, flat steel disc.

3. Procedure.

3.1 Sample preperation.

3.1.1 Pulverize cast, molded, and cured extruded material in a ball or Wally mill.

3.1.2 Store materials in desiccators for a least 24 hours prior to test. (Materials are normally tested dry, in either the powdered or the granular state.)

3.1.3 Place approximately 25 mg powdered sample in the sample holder.

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3.1.4 Place electrical insulating tape, 0.19 mm thick, over the sample opening to confine the sample.

3.2 Electrode preparation.

3.2.1 Replace needle after any trial in which there is evidence of a reaction, whenever a test of a new material is started, or when any other condition dictates.

3.2.2 Set electrode spacing (gap) to 0.18 mm (0.007 inch)).

3.2.3 Wipe needle with a “kimwipe”, or equivalent absorbent paper, after every trial.

3.2.4 Clean and polish steel sample holder in which there is evidence of reaction or when any other condition dictates. {Cleaning is done using first, No. 400, then No. 600 emery cloth, and finally, polishing with crocus cloth.)

3.3 Test procedure.

3.3.1 Connect the 0.02 microfarad capacitor to the discharge circuit. The starting capacitor is usually the largest value unless a more efficient value based on experience is known.

3.3.2 Raise the upper electrode (needle) to cocked position by means of remote handle.

3.3.3 place sample holder containing the test sample on the base electrode with the powder directly under the needle.

3.3.4 Close the door.

3.3.5 Turn on power supply and adjust voltage control to 5 kv.

3.3.6 Activate reset switch to charge the capacitor.

3.3.7 Adjust power supply until the electrostatic voltmeter reads 5 EV.

3.3.8 Pull release rod to release the approaching electrode. The charged electrode will rapidly move downwards to the present gap distance. The needle will puncture the tape, penetrate the sample material, discharge through the interstices of the material, and rise again to its initial position.

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3.3.9 Record reaction or no reaction. A reaction is indicated by a severed tape, whereas no reaction is evidenced by a punctured but otherwise intact tape.

3.3.10 Repeat the procedure until no reaction is obtained in 20 consecutive trials. If a reaction is obtained, the energy is reduced by decreasing the potential on the capacitor in 500 v increments and the above procedure repeated. The voltage is reduced until the charging voltage is 2500 v and then the next lower capacitor is selected.

3.3.11 Report results “NO reaction at _____ Joule" according to the following table.

Capacitance (microfarad)	Voltage (kv)	Approximately delivered energy (J)
0.02	5.0	0.250
	4.5	0.200
	4.0	0.160
	3.5	0.122
	3.0	0.090
	2.5	0.063
0.01	3.0	0.045
	2.5	0.031
0.005	3.0	0.023
	2.5	0.016
0.0025	3.0	0.0113
	2.5	0.0078
0.001	3.5	0.0061
	3.0	0.0045
	2.5	0.0031
0.0005	3.0	0.0023
	2.5	0.0016
0.0002	3.5	0.0012
	3.0	0.0009
	2.5	0.0006

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METHOD T 513.1

LARGE SCALE GAP TEST

(U.S. Naval Ordnance Laboratory Method)

1. Scope

This test is used to determine the sensitivity of an explosive material to shock.

2. Apparatus.

a. Two pentolite pellets, each 2 in. diameter x 1 in. long pressed to density of 1.51 g\cc (Donor) .

b. Cellulose acetate discs (“cards”) 2 in. diameter by 0.010 in. thick.

c. pellets of polymethyl methacrylate (PMMC), 2 in. diameter by 0.50 in. thick, and 2 in. diameter by 1.0 in. thick.

d. Witness plage - 4 in. square sheet of commercial cold rolled mild steel, 0.375 in. thick.

e. No 6 blasting cap, loaded with 0.87 g of PETN and 0.34 g of diazodinitrophenol.

f. Wooden block, 2 in. diameter by 1 in. thick.

g. Seamless pipe, of cold drawn, slid steel, 1.44 in. inside diameter by 1.875 in. outside diameter by 5.50 in. long.

3. Procedure.

3.1 Preparation of Apparatus

3.1.1 Figure 1 shows the assembly of the test elements to be used which consist of an explosive donor, an alternating spacer, an explosive acceptor, and a metal witness plate.

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3.1.2 In order to obtain variable gaps between donor and acceptor, both types of spacers (cards and PMMC pellets) may be used. Combinations of the 1.0. 0.50 and O.O1O in. thickness produce any desired ga) in steps of 0.010 inch.

3.1.3 Place two wood blocks on the ground to support the assembly.

3.1.4 Load the test explosive into the seamless pipe. The explosive may be cast or pressed directly into the steel tube or it may be prepared as a bare billet and machined to slip into the tube with a close fit. The density of the acceptor shall be the normal loading density (as specified for weapon filling).

3.1.5 Slip the detonator into the central hole of the wooden block.

3.1.6 Place the detonator block, pentolite pellets, gap cards and acceptor charge (loaded in its steel tube) inside the paperboard tube in that order, from base to top.

3.1.7 Insert a paperboard spacer tube between the acceptor charge and the outside tube to position the acceptor centrally. Check that the spacer projects approximately 1/16 in. beyond the top end of the acceptor.

3.1.8 Place the steel witness plate on top of the upper spacer tube so that there is an air gap of 1/16 in. between acceptor and witness plate and so that the acceptor is approximately centered beneath the plate.

3.2 Test Procedure.

3.2.1 In each test, if the steel witness plate contains a hole it is considered a "go". If the plate is only distorted it is considered a “no-go”. For the witness plate used, the test explosive must react to a pressure of over 60 kbar to perforate the plate. (NOTE: The detonation pressures of most condensed explosives are two to seven times this value.)

3.2.2 If the sensitiveness of the test explosive is unknown, perform the first test without attenuating spacers (zero gap). If detonation (a “go”) occurs, perform the next test with a gap of 0.08 in. Thereafter, double the gap width with each additional test until the test explosive does not detonate (“no-go”).

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3.2.3 After the first “no-go”, perform each successive shot in a down-and-up manner, with a gap halfway between the closest “go” and “no-go” values. continue this procedure until a “go” and “no-go” have been obtained which differ by only 0.010 in. gap.

3.2.4 If the sensitiveness of the test explosive has been measured or can be predicted, perform the first test with a gap greater than zero. Perform succesive tests in the same up and down manner as before, depending on the results of each previous shot.

3.2.5 It is usually possible to complete the testing in a total of 8 to 12 shots (including the last set of shots described below).

3.3 Evaluation and Calculations.

3.3.1 Report the test result as the gap in inches which corresponds to a 50% probability of detonation of the test explosive (The 50% point may also be reported as a “number of cards”, where the value of a “card” is a gap of 0.010 in.).

3.3.2 Determine the 50% probability point as follows. When testing has proceeded through that described in paragraph 3.2.3, designate the gap size corresponding to the “go” as N, and the gap corresponding to the “no-go” as N+1.

3.3.3 A basic test pattern of four shots is established by making one additional shot at both N and N+1 gaps. Table I shows the four possible results of this pattern and their interpretation. If the pattern of results follow pattern designation III of IV then perform two additional test firings to determine the 50% point.

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3.3.4 Typical test results are given for 5 common explosives in the table below:

Explosive	Pressed Density g/cm^3	50% Point - In.
RDX	1.64	3.23
Tetryl*	1.62	2.61
Comp B	1.66	2.38
TNT	1.60	1.83
Explosive D	1.59	1.50

*Contained 0.5% graphite, by weight

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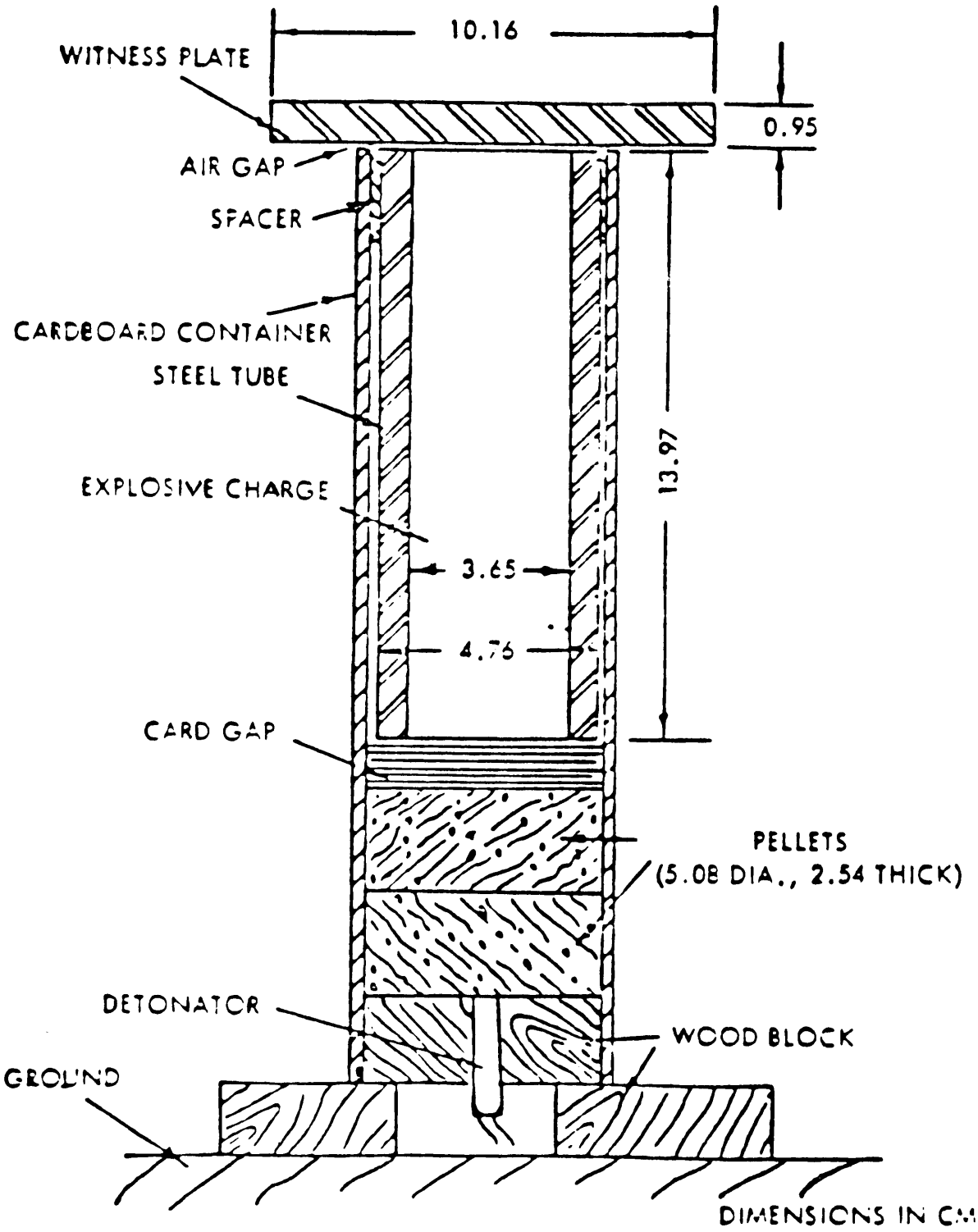


FIG 1. GAP TEST ASSEMBLY

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TABLE 1

DETERMINATION OF 50% POINT FOR VARIOUS TEST CONFIGURATIONS
(After Detonation on First Test at Zero Gap)

N = an integer (in units of 0.010 in.)
\+ = detonation
\- = no detonation
... = no tests needed

BASIC TEST PATTERN*				RESULTS OF SUPPLEMENTARY TESTS*		
	Size of Gap		Supplementary Tests Required to Establish 50% Point	Size of Gap		
Designation	N	N+l		N-l	N+2	50% Point
I	++	--	NONE	...	...	N + 1/2
II	+-	-+	NONE	...	...	N + 1/2
III	+-	--	Two tests at N-1	--		N - 1
				++	...	N
				-+	...	N - 1/2
IV	++	-+	Two tests at N+2	...	--	N + 1
				...	++	N + 2
				...	-+	N + 1-1/2

* For a particular number of cards the order in which the results are obtained is immaterial.

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METHOD T 514.1

AUTOIGNITION TEMPERATURE

1. Scope.

1.1 This methos is used to determine the lowest temperature at which self-heating of an explosive occurs followed by ignition.

2. Apparatus.

a. Deltatherm III Differential Thermal Analysis Accessory or equivalent DTA apparatus which is precisely calibrated for temperature.

b. Mosely 7100B two pen time-base recorder or equivalent

3. Procedure.

3.1 Preparation of Sample.

3.1.1 Use a sample size which will produce an isosceles triangle of not too great an angle. Too steep an angle (80-90°) is not acceptable (too large a sample); and too gradual an angle (0-35°) is not acceptable (too small a sample). Start with .2 mg for 10°/minute and 2 mg for 2.5°/minute.

3.2 Test Procedure.

3.2.1 Follow general procedure for differential thermal analysis, ASTM E474 or manufacturer’s instructions.

3.2.2 Place the sample and reference material into aluminum or steel cup (with size of up to 4.8mm) and place into thermo cups (steel cups to which an alumel and a chromel wire are welded) or similar thermal measuring receptacles.

3.2.3 close experimental area and set heating rate. The rates to be used shall be 10,5,2 l\2 and 1°C per minute respectively.

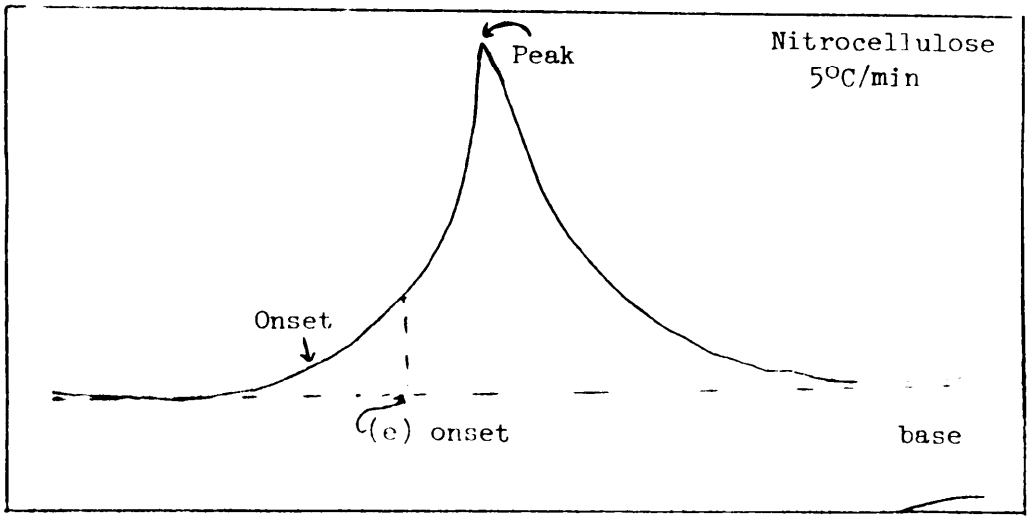
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3.2.4 Set limits on recorder L(paper speed, sensitivity) and instrument sensitivity. One pen shall record the temperature of the reference material; the other pen shall record the T of the 2 cups at a space of 1/10 in. ahead of the temperature pen for the Deltatherm III apparatus.

3.2.5 To minimize the autocatalytic activity, pass a constant flow of argon or nitrogen across the sample holder while heating the sample at a constant rate. (30-50 milliliters/minute)

3.3 Calculations.

3.3.1 Determine "e^n extrapolated onset and peak temperature by using the following guideline:



3.3.2 Graph $Ø/T^2$ VS. T^{-1}, where Ø = heating rate, C/min, T = peak temperature, to obtain the activation energy (slope) from 4 heating rate tests.

3.3.3 The following relationship is used to solve for the necessary variables.

$$E\phi/RT^2 = Ae^{-E/RT} = 0.05/min$$

$$\text{or } E\phi/RT^2 = \exp -(AE/RT) = 0.05/min$$

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3.3.4 Obtain A, the preexponential factor, at each peak temperature (at the four heating rates) and use the average of these values for the equation. This is done by solving for A in the above equation. (See attached excerpt from Thermochimica Acts, 14 (1976), 183-199).

3.3.5 Set $E\emptyset/RT^2$ equal to 0.05 and solve for the auto ignition temperature from T in the exponent.

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TABLE 2
CALCULATION OF ACTIVATION ENERGY AND
AUTOIGNITION TEMPERATURE FOR TNT (CONFINED) KNK 11-270

Temperature (°C)	Temperature (K)	Rate = ϕ (°C min^{-1})	$1/T + 10^3$ (K)	$\phi/T^2 \times 10^{-5}$
Temperature at the peak of the exotherm				
333	606	22.0	1.650	5.847
315	588	10.7	1.701	3.058
296	569	5.3	1.757	1.637
287	560	2.6	1.786	0.829
Temperature at the beginning of the slope to the exotherm peak				
326	599	22.0	1.666	5.988
309	582	10.7	1.718	3.121
292	565	5.3	1.770	1.660
285	558	2.6	1.792	0.835

$$\text{slope} = \frac{E}{\ln R} = \frac{\log (\phi/T^2)_1 - (\phi/T^2)_2}{1/T_1 - 1/T_2}$$

E_p = 24 200 cal mol^{-1}; E_s = 24 900 cal mol^{-1}.
E_p = activation energy at peak; E_s = activation energy at slope; ϕ = constant heating rate; T = temperature (K); R = gas constant; A = frequency factor (sec^{-1}).

Peak temp. (K)	$\phi/T^2 \times 10^{-5}$	$\frac{E\phi}{RT^2}$	$e^{-E/RT} \times 10^{-9}$	$A \times 10^8$
606	5.991	0.72965	1.865	3.912
588	3.095	0.37695	1.013	3.721
568	1.642	0.1998	0.488	4.094
560	0.829	0.10096	0.358	(2.821)
				average (3) = 3.90×10^8
Slope temp. (K)				
599	6.131	0.76799	0.830	9.253
582	3.159	0.39571	0.451	8.774
565	1.660	0.20794	0.235	8.849
558	0.834	0.10447	0.178	(5.869)
				average (3) = 8.96×10^8

$$\frac{E\phi}{RT^2} = Ae^{-E/RT}$$

as $\phi \to 0$
$E\phi/RT^2 \to 0.1$

$$\log 0.1 = \log 3.90 \times 10^8 - \frac{E}{RT \times 2.303}$$

T = 552K
279.0°C peak temperature

$$\log 0.1 = \log 8.96 \times 10^8 \frac{E}{RT \times 2.303}$$

T = 547.6 K
= 274.6°C autoignition temperature confined.

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METHOD T 515.1

EXPLOSION TEMPERATURE TEST

(U.S. Army ARDEC Laboratory Method)

1. Scope

This thermal test is used to determine the 5-second time-to-explosion.

2. Apparatus.

Apparatus used for this test includes: a barricaded press with a die for loading 40 milligrams of explosive sample into a No. 8 blasting cap; a crimper; a furnace with a molten woods bath and shield; associated temperature measurement and controls; and a time interval recorder.

3. Procedure.

3.1 Preparation of Apparatus.

3.1.1 Check press operation and set loading force for 230 lbs. with Dillon gauge.

3.1.2 Place an empty blasting cap in the loading fixture (die).

3.1.3 Weigh out the proper amount of explosive (40 mg), or use a previously calibrated conductive volumetric scoop, and pour into cap using loading funnel.

3.1.4 Insert a copper gas check, open side up, into cap and manually push the gas check down to bottom of cap with rod.

3.1.5 Put 30 mg of dried talc in cap, insert second gas check, and repeat paragraph 3.1.4.

3.1.6 Close barricade door and apply 6000 psi pressure (230 lbs.loading force) for ten seconds.

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3.1.7 Remove die from press. Behind a shield, remove cap from die with shielded tweezers and transfer cap to crimper.

3.1.8 Push cap into crimper with shielded tweezers and hold against stop.

3.1.9 Slowly lower handle of crimper until resistance is felt, then remove tweezers and crimp until stop is reached.

3.1.10 Raise handle of crimper and remove loaded cap with tweezers and place in nonpropagating barricade.

3.2 Test Procedure.

3.2.1 Put a piece of masking tape around top of each cap.

3.2.2 Heat Woods metal bath to highest desired temperature. Stir bath with steel rod and wait at least 5 minutes until temperature is uniform.

3.2.3 Rotate sample holder to loading position.

3.2.4 Reset timer to zero. Check start and stop signals, then reset zero.

3.2.5 Insert loaded cap in hole in fixture using shielded tweezers, and press into position. (Tape should hold cap firmly up, if not, add or remove tape as necessary) .

3.2.6 Grasp handle and rotate fixture to allow sample to be immersed into bath.

3.2.7 When an explosion occurs, record time, and temperature, and using handle, lift fixture and rotate back to the loading position.

3.2.8 If an explosion does not occur within two minutes, remove sample from bath as described in paragraph 3.2.7 and allow sample to cool for at least 1 minute. Then destroy sample; using shielded tweezers, place the cap behind a hood safety shield in a hole in a shielded steel block on a hot plate or in a furnace at 400°C temperature to destroy the explosive at end of test, allow sample to cool and then dispose of in explosive waste can (underwater).

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3.2.9 After all samples to be fired at the temperature of the bath have been fired, lower the temperature setting on the controller to the next temperature desired. Add cold metal, then stir thoroughly and check temperature. If still too hot, and more cold metal and stir again. Remove excess metal with a spoon and pour onto a steel plate to cool for future use. As long as the temperature is known at the time the sample is tested, the exact temperature set on the controller is not required.

3.2.10 Perform successive tests in the same manner to obtain time-to-explosion temperature values over a time range of 0.5 to 9 seconds.

3.3 Calculations.

3.3.1 Report the test result as the Temperature in degrees centigrade which corresponds to the 5-second-explosion temperature.

3.3.2 Determine the 5-second-explosion temperature as follows. Plot the data on semilog paper, the logarithm of the time-to-explosion versus the reciprocal of the absolute temperature. This results in a straight line. A linear least square curve fit program should be used to determine the temperature value for the 5-second time-to-explosion.

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